

Introduction to GAMs with mgcv

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Overview

A **Generalized Additive Model (GAM)** extends the linear model by replacing linear predictor terms with smooth functions of covariates. The model takes the form:

$$g(\mu_i) = \beta_0 + f_1(x_{1i}) + f_2(x_{2i}) + \cdots + f_p(x_{pi})$$

where g is a link function, $\mu_i = \mathbb{E}(y_i|\mathbf{x}_i)$, and each f_j is a smooth function estimated from the data using penalized regression splines.

The smooth functions are represented as linear combinations of basis functions:

$$f_j(x) = \sum_{k=1}^{K_j} \beta_{jk} b_{jk}(x)$$

Smoothness is controlled by a wiggleness penalty, and the smoothing parameter λ is estimated automatically (e.g., by REML).

This vignette demonstrates fitting a simple GAM to simulated data using **mgcv** in R.

Setup

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

```
library(gratia)
library(ggplot2)
```

Loading data

We load $n = 200$ observations generated from a sine curve with Gaussian noise:

$$y_i = \sin(2\pi x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 0.3^2)$$

```
dat <- read.csv("../data.csv")
x <- dat$x
y <- dat$y
n <- nrow(dat)
df <- data.frame(x = x, y = y)
head(df)
```

	x	y
1	0.0002388966	0.36179064
2	0.0013808436	0.32210131
3	0.0015705542	-0.29109467
4	0.0022729661	0.56882555
5	0.0039483388	-0.17522642
6	0.0073341469	0.07771964

Fitting a GAM

We fit a GAM with a cubic regression spline (`bs = "cr"`) smooth of `x` using 15 basis functions:

```
m <- gam(y ~ s(x, k = 15, bs = "cr"), data = df, method = "REML")
summary(m)
```

Family: gaussian

Link function: identity

Formula:

`y ~ s(x, k = 15, bs = "cr")`

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.1002	0.0205	-4.888	2.15e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(x)	7.722	9.369	122.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.852 Deviance explained = 85.8%

-REML = 49.744 Scale est. = 0.08403 n = 200

The model summary shows parametric coefficients, smooth term EDF, deviance explained, and scale estimate.

Coefficients and deviance explained

```
cat("Number of observations:", nobs(m), "\n")
```

Number of observations: 200

```
cat("EDF per smooth:      ", round(pen.edf(m), 2), "\n")
```

EDF per smooth: 7.72

```
cat("Deviance explained:   ", round(summary(m)$dev.expl * 100, 1), "%\n")
```

Deviance explained: 85.8 %

```
cat("Scale estimate:      ", round(m$scale, 4), "\n")
```

Scale estimate: 0.084

Smooth estimates

The `gratia::smooth_estimates()` function evaluates the estimated smooth on a regular grid and returns pointwise standard errors:

```
se <- smooth_estimates(m, n = 200)
head(se)
```

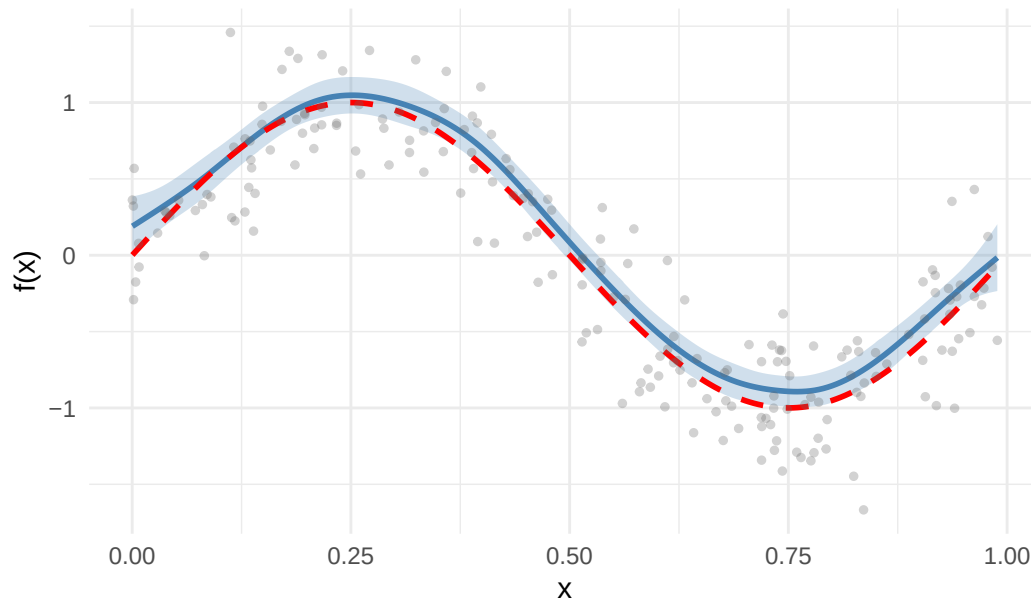
```
# A tibble: 6 x 6
  .smooth .type .by .estimate .se      x
  <chr>   <chr> <chr>   <dbl> <dbl> <dbl>
1 s(x)    CRS   <NA>    0.190 0.0976 0.000239
2 s(x)    CRS   <NA>    0.208 0.0915 0.00521
3 s(x)    CRS   <NA>    0.226 0.0859 0.0102
4 s(x)    CRS   <NA>    0.244 0.0809 0.0151
5 s(x)    CRS   <NA>    0.263 0.0765 0.0201
6 s(x)    CRS   <NA>    0.281 0.0729 0.0251
```

We plot the smooth estimate with ± 2 SE confidence bands, overlaying the true function:

```
ggplot(se, aes(x = x, y = .estimate)) +
  geom_ribbon(aes(ymin = .estimate - 2 * .se, ymax = .estimate + 2 * .se),
    alpha = 0.25, fill = "steelblue") +
  geom_line(linewidth = 1, colour = "steelblue") +
  geom_line(aes(y = sin(2 * pi * x)), linetype = "dashed",
    linewidth = 1, colour = "red") +
```

```
geom_point(data = df, aes(x = x, y = y), alpha = 0.3, size = 1, colour = "grey40") +
labs(x = "x", y = "f(x)", title = "Estimated smooth with confidence band") +
theme_minimal()
```

Estimated smooth with confidence band



Prediction at new points

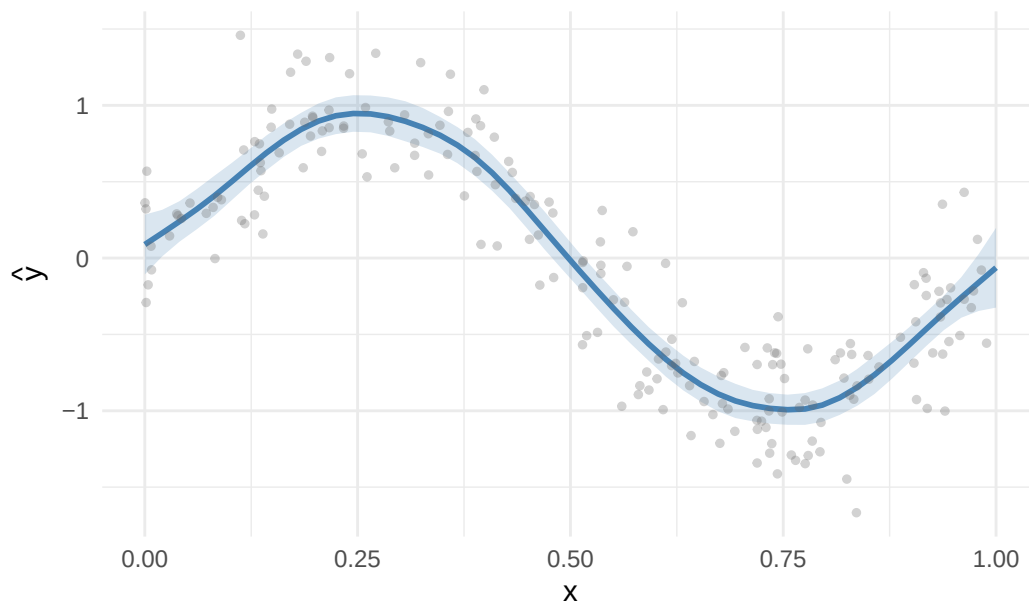
We can predict at new covariate values with standard errors:

```
x_new <- data.frame(x = seq(0, 1, length.out = 50))
pred <- predict(m, newdata = x_new, type = "response", se.fit = TRUE)

x_new$fit <- pred$fit
x_new$se <- pred$se.fit

ggplot(x_new, aes(x = x, y = fit)) +
  geom_ribbon(aes(ymin = fit - 2 * se, ymax = fit + 2 * se),
            alpha = 0.2, fill = "steelblue") +
  geom_line(linewidth = 1, colour = "steelblue") +
  geom_point(data = df, aes(x = x, y = y), alpha = 0.3, size = 1, colour = "grey40") +
  labs(x = "x", y = expression(hat(y)), title = "Predictions at new data") +
  theme_minimal()
```

Predictions at new data



Smoothing parameters and REML

`mgcv` estimates the smoothing parameter λ by minimizing the Restricted Maximum Likelihood (REML) criterion by default. REML is generally preferred over GCV because it:

- Avoids the tendency to undersmooth
- Provides more stable estimates
- Has a natural Bayesian interpretation

```
cat("Estimation method:", m$method, "\n")
```

Estimation method: REML

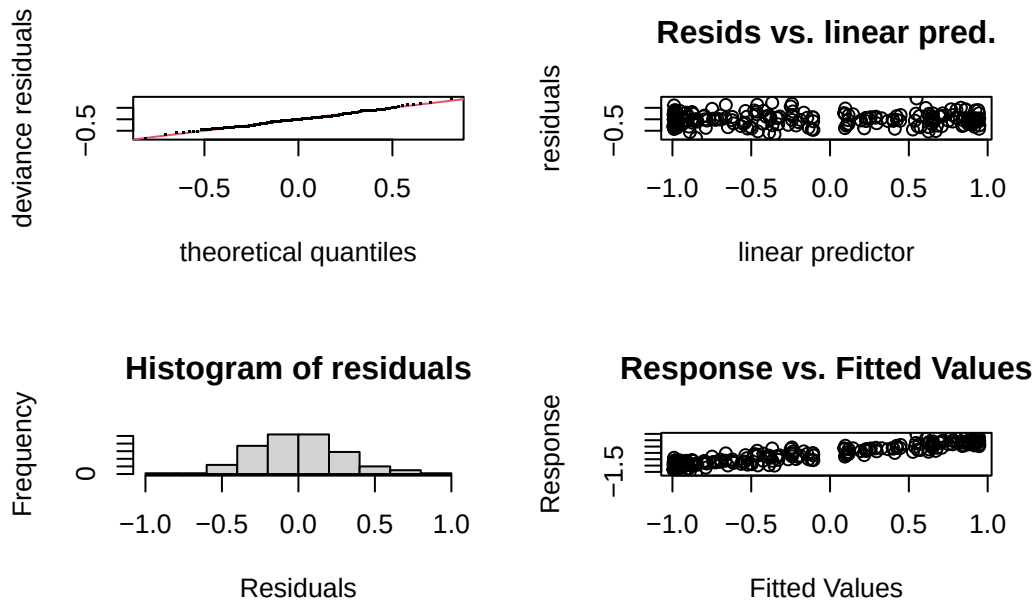
```
cat("Smoothing parameter (log):", log(m$sp), "\n")
```

Smoothing parameter (log): 4.278883

Model diagnostics

The `gam.check()` function provides diagnostic plots and basis dimension checks:

```
par(mfrow = c(2, 2))
gam.check(m)
```



Method: REML Optimizer: outer newton
 full convergence after 4 iterations.
 Gradient range [-6.776588e-10,3.284431e-10]
 (score 49.74438 & scale 0.08402973).
 Hessian positive definite, eigenvalue range [3.328508,99.11809].
 Model rank = 15 / 15

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(x)	14.00	7.72	1.09	0.9

```
par(mfrow = c(1, 1))
```

Summary

In this vignette we:

1. Simulated data from a sine curve with Gaussian noise
2. Fitted a GAM using a cubic regression spline with $k = 15$ basis functions
3. Examined the model summary, EDF, and deviance explained
4. Visualized the smooth estimate with confidence bands using `gratia`
5. Predicted at new data points with standard errors
6. Discussed REML-based smoothing parameter estimation

The next vignette compares different smooth basis types.